TRANSFORMER MODEL TO RELATIONAL MODEL

TRANSFORM - ER MODEL TO RELATIONAL MODEL

- Both ER-Model and Relational Model are abstract logical representation of real-world enterprises. Because the two models implies the similar design principles, we can convert **ER design into Relational design**.
- Converting a DB representation from an ER diagram to a table format is the way we arrive at Relational DB-design from an ER diagram.

ER DIAGRAM NOTATIONS TO RELATIONS:

- **Strong Entity**
  - Becomes an individual table with entity name, attributes becomes columns of the relation.
  - Entity’s Primary Key (PK) is used as Relation’s PK.
  - FK are added to establish relationships with other relations.
- **Weak Entity**
  - A table is formed with all the attributes of the entity.
  - PK of its corresponding Strong Entity will be added as FK.
  - PK of the relation will be a composite PK, {FK + Partial discriminator Key}.

- **Single Valued Attributes**
  - Represented as columns directly in the tables/relations.
- **Composite Attributes**
  - Handled by creating a separate attribute itself in the original relation for each composite attribute.
  - e.g., Address: {street-name, house-no}, is a composite attribute in customer relation, we add address-streetname and address-house-name as new columns in the attribute and ignore "address" as an attribute.

- **Multivalued Attributes**
  - New tables (named as original attribute name) are created for each multivalued attribute.
  - PK of the entity is used as column FK in the new table.
  - Multivalued attribute's similar name is added as a column to define multiple values.
  - PK of the new table would be {FK + multivalued name}.
  - e.g., For Strong entity Employee, dependent-name is a multivalued attribute.
    - New table named dependent-name will be formed with columns emp-id, and dname.
    - PK: {emp-id, name}
    - FK: {emp-id}
- **Derived Attributes**: Not considered in the tables.

RULE-1

1. Every **strong entity** will be a relation /table. Every single valued attribute will be the column. For Composite attribute add its simple attributes as columns. Leave the multi valued and derived attributes. Primary attribute has to be underlined.

2. Eg: Student Entity

RULE-2

- **Weak entity** will be a table. Add its attribute as columns. Also create a foreign key, refer to the primary key of the strong entity.
- Combination of FK and other key will be treated as key.
- In case of weak entities, we have to check the on delete cascade.

- Eg: Employee and dependents

RULE-3

- In case of 1 to 1 Relationship
- For one side total participation, create two tables. Include foreign key in the entity with total participation.
- Eg: Employee manages Department

- For both side total participation, We can create two table, and can keep foreign key on either side.
- Eg: Employee holds a bank account
- We can even merge two tables.

RULE-4

- In case of N to 1 Relationship,
- Create two tables and link by using foreign key in N side.
- Employee works for a department

RULE-5

- In case of M to N relationship.
- Employee works in a project
- Student Enrolled in a course

- Create two table for entity and one more table to represent the relation. Add two foreign keys, referring to primary key of each entity. Also, if the relation has any attribute add it to the newly formed table.

RULE-6

- For Multi-valued attribute, create a sperate table for it. Add a foreign key referring to the primary key of the entity.
- Eg Student entity having phone number
- Eg Customer entity having address

RULE-7

- If there are more than two entities are there in a relationship
- Create a table for the relationship, Add the primary key of all entities in it as foreign key.
- If the relationship has any attribute add that to the table.

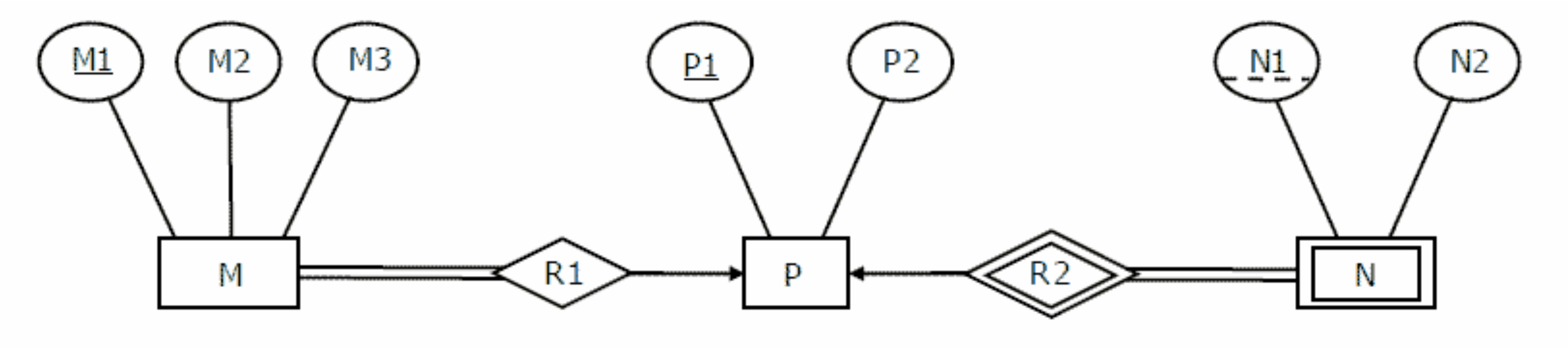
- Consider the entities ‘hotel room’, and ‘person’ with a many to many relationship ‘lodging’ as shown in the figure. If we wish to store information about the rent payment to be made by person (s) occupying different hotel rooms, then this information should appear as an attribute of

  a. Hotel Room
  b. Person
  c. Lodging
  d. None

Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

- A Rectangles with double/bold border
- B Diamonds with double/bold border
- C Ovals that contain underlined identifiers
- D Ovals with double/bold border

- Given the basic ER and relational models, which of the following is INCORRECT?
    a. An attribute of an entity can have more than one value
    b. An attribute of an entity can be composite
    c. In a row of a relational table, an attribute can have more than one value
    d. In a row of a relational table, an attribute can have exactly one value or a NULL value



- The minimum number of tables needed to represent M, N, P, R1, R2 is
    a. 2
    b. 3
    c. 4
    d. 5

- The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on-delete cascade.

  The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple (2,4) is deleted is:

  a. (3,4) and (6,4)
  b. (5,2) and (7,2)
  c. (5,2), (7,2) and (9,5)
  d. (3,4), (4,3) and (6,4)

A	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

- Let E1 and E2 be two entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?
    a. 2
    b. 3
    c. 4
    d. 5

  If E1 has two attributes A and B(A primary key) E has three attributes C,D (C primary key) Draw the relational model.

- Let R(a,b,c) and S(d,e,f) be two relations in which d is the foreign key of S that refers to the primary key of R. Consider the following four operations R and S
    1. Insert into R
    2. Insert into S
    3. Delete from R
    4. Delete from S
- Which of the following can cause violation of the referential integrity constraint above?
    a. None of (1), (2), (3) or (4) can cause its violation
    b. All (1), (2), (3) and (4) can cause its violation
    c. Both (1) and (4) can cause its violation
    d. Both (2) and (3) can cause its violation

- In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in R
- Which one of the following is true about R?
    a. Every entity in E1 is associated with exactly one entity in E2
    b. Some entity in E1 is associated with more than one entity in E2
    c. Every entity in E2 is associated with exactly one entity in E1
    d. Every entity in E2 is associated with at most one entity in E1

Which of the following is **NOT** a superkey in a relational schema with attributes V, W, X, Y, Z and primary key VY?

A. $VXYZ$

B. $VWXZ$

C. $VWXY$

D. $VWXYZ$

StudentID	StudentName	StudentEmail	StudentAge	CPI
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

- Given the STUDENTS relation as shown For (StudentName, StudentAge) to be the key for this instance, the value X should not be equal to

  (A) 18

  (B) 19

- Consider a relational table with a single record for each registered student with the following attributes.
  - ❖ *Registration_Num*: Unique registration number of each registered student
  - ❖ *UID*: Unique identity number, unique at the national level for each citizen
  - ❖ *BankAccount_Num*: Unique account number at the bank. A student can have multiple accounts or join accounts.
  - ❖ *Name*: Name of the student
  - ❖ *Hostel_Room*: Room number of the hostel

Which one of the following option is **INCORRECT**?

(A) BankAccount_Num is candidate key
(B) Registration_Num can be a primary key
(C) UID is candidate key if all students are from the same country
(D) If S is a superkey such that S∩UID is NULL then S∪UID is also a superkey